

Math Shortcuts (`math.sty`)

Jambro Beamer Theme

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Math Shortcuts

(`math.sty`)

Math shortcuts: overview

- ▶ Jambro ships with `math.sty`, a companion file of compact math-mode macros adapted from `pmichaillat/math-for-macro`
 - * Brackets and norms: `\bp`, `\bs`, `\abs`, `\norm`, ...
 - * Statistical operators: `\E`, `\var`, `\cov`, `\corr`, `\sd`, `\P`
 - * Calculus: `\od`, `\pd`, `\oe`, `\pe`
 - * Greek (`\a`, `\b`, ...), blackboard ($\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$), and calligraphic ($\mathcal{A}$, $\mathcal{B}$, ..., $\mathcal{Z}$)
- ▶ Load it once in the preamble, after the theme: `\usetheme{jambro}` then `\input{math.sty}`
- ▶ Caveat: Greek shortcuts shadow TeX text-accent macros (`\b`, `\c`, `\d`, `\u`, `\t`, `\o`); fine for English decks, watch out if you also typeset diacritics

Brackets and delimiters

► Auto-sized brackets that wrap their argument

- * $\backslash bp\{x\} \rightarrow (x)$, $\backslash bs\{x\} \rightarrow [x]$, $\backslash bc\{x\} \rightarrow \{x\}$, $\backslash ba\{x\} \rightarrow \langle x \rangle$
- * $\backslash of\{x\} \rightarrow (x)$ for function arguments

► Norms, absolute values, floor, ceiling

- * $|x|$, $\|v\|$, $\lfloor x \rfloor$, $\lceil x \rceil$ via $\backslash abs$, $\backslash norm$, $\backslash floor$, $\backslash ceil$
- * Non-stretching variants $\backslash absx$, $\backslash normx$, $\backslash floorx$, $\backslash ceilx$ for inline text

► Brackets scale to fit tall arguments

$$\left| \frac{1}{1+x^2} \right| \quad \left\| \sum_{i=1}^n x_i e_i \right\| \quad \left\lfloor \frac{n}{2} \right\rfloor \quad \langle u, v \rangle$$

Operators and statistical functions

- Expectation, probability, and moments take an optional subscript and a grouped argument

- * $\backslash E\{X\} \rightarrow \mathbb{E}(X)$, $\backslash E[t]\{X_{t+1}\} \rightarrow \mathbb{E}_t(X_{t+1})$

- * $\mathbb{P}(A | B)$, $\text{var}_t(X_{t+1})$, $\text{cov}(X, Y)$, $\text{corr}(X, Y)$, $\text{sd}(X)$, $\text{se}(\hat{\beta})$

- Logarithm and exponential become parens-aware when given an argument

- * Bare $\backslash \ln \rightarrow \ln$; $\backslash \ln\{1+x\} \rightarrow \ln(1+x)$; $\backslash \exp\{-r \ t\} \rightarrow \exp(-rt)$

- Indexed extrema: $\max_i(x_i)$, $\min_i(x_i)$, $\sup_t(X_t)$, $\inf_{\theta}(f(\theta))$

- $\backslash \operatorname{argmax}$, $\backslash \operatorname{argmin}$, $\backslash \operatorname{ind}(\text{indicator})$

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y), \quad \text{var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2, \quad \mathbb{1}(y > 0)$$

Derivatives and elasticities

- ▶ Ordinary derivatives: $\frac{dy}{dx}$ and higher orders via $\frac{d^2y}{dx^2}$

$$\frac{dy}{dx} \quad \frac{d^2y}{dx^2} \quad dy/dx \text{ (text variant)}$$

- ▶ Partial derivatives, with optional evaluation point

$$\frac{\partial u}{\partial x} \quad \frac{\partial^2 u}{\partial x^2} \quad \frac{\partial u}{\partial x} \Big|_{x=0}$$

- ▶ Ordinary and partial elasticities

$$\frac{d \ln(Y)}{d \ln(P)} = \frac{d \ln(Y)}{d \ln(P)} \quad \frac{\partial \ln(Y)}{\partial \ln(P)} \Big|_{P_0}$$

Logical, statistical, and accent shortcuts

► Logical relations: $X \Rightarrow Y$ via `\implies`, $X \Leftrightarrow Y$ via `\iff`

► Statistical convergence

$$* X_n \overset{iid}{\sim} F (\backslash iid), X_n \overset{as}{\rightarrow} X (\backslash asto), X_n \overset{p}{\rightarrow} X (\backslash pto), X_n \overset{d}{\rightarrow} X (\backslash dto)$$

► Math accents

$$* \bar{x} (\backslash ubar), \underline{x} (\backslash ul), \overrightarrow{x} (\backslash oa), \overline{x} (\backslash ol), \hat{x} (\backslash wh), \tilde{x} (\backslash wt)$$

► Putting it together

$$\hat{\beta}_n \overset{p}{\rightarrow} \beta, \quad \sqrt{n} (\hat{\beta}_n - \beta) \overset{d}{\rightarrow} \mathcal{N}(0, \sigma^2)$$

Greek, blackboard, and calligraphic letters

- ▶ Lowercase Greek collapse to a single character

* `\a` α , `\b` β , `\g` γ , `\d` δ , `\e` ϵ , `\z` ζ , `\h` η , `\t` θ , `\k` κ , `\l` λ , `\m` μ , `\n` ν , `\r` ρ , `\s` σ , `\f` ϕ , `\p` ψ , `\o` ω

* Capitals: `\D` Δ , `\G` Γ , `\L` Λ , `\S` Σ , `\T` Θ , `\F` Φ , `\O` Ω , `\X` Ξ , `\U` Υ

- ▶ Blackboard: `\R` $\mathbb{R}$, `\N` $\mathbb{N}$, `\Z` $\mathbb{Z}$, `\Q` $\mathbb{Q}$, `\C` $\mathbb{C}$, `\V` $\mathbb{V}$

- ▶ Calligraphic: `\Ac` $\mathcal{A}$, `\Bc` $\mathcal{B}$, `\Fc` $\mathcal{F}$, `\Nc` $\mathcal{N}$, ..., `\Zc` $\mathcal{Z}$

- ▶ In combination, equations stay short

$$y_t = \alpha + \beta x_t + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2), \quad \mathbb{E}_t(x_{t+1} | \mathcal{F}_t) \in \mathbb{R}$$

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